

GEOOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Ottertail Power Company 215 S. Cascade St Fergus Falls, MN 56537 ATTN: Brandon Johnson, CEM	2025-10-22	QU-0217
		Reference
		US MN Butler Machinery

Understanding of Project

OTPCO is supporting a customer developing a new machine shop facility in Mapleton, MN. The project includes approximately 40,000 square feet of shop space and 12,000 square feet of office space. Butler Machinery is interested in evaluating the potential use of a geothermal heat pump system for the building. To assist in this evaluation, OPCO has requested a feasibility analysis to determine the viability and benefits of implementing a geothermal solution for this project.

Scope of Work

Feasibility

- Develop a detailed hourly energy model of the proposed building.
 - The model may include several iterations of the proposed building to determine the ventilation strategy, glass specifications and other potential energy efficiency measures that will result in the most cost-effective ground heat exchanger for the facility.
- Ground Loop Design (GLD) model to size the ground heat exchanger (GHX)
- Review of the geology of the site to determine the thermal properties of the soil and the most cost-effective drilling depth for potential drilling contractors in the area.
- Provide size and cost estimates for either a horizontal and vertical GHX options to support a life-cycle cost analysis (LCCA), including operational costs and a financial evaluation of ROI, NPV, and simple payback.

Information Required

- Detailed construction drawings and specifications for the proposed building, including:
 - Insulation values of walls, floor, and roof.
 - Window specifications (solar heat gain coefficient (SHGC) and U-value).
 - Estimated occupancy schedule of the building.
 - Proposed mechanical system for the building, including ventilation requirements and specifications.

- Electrical and lighting drawings and specifications for the building.
- Site plan indicating land area available for construction of the GHX.
- Geotechnical report.

Deliverables

- Preliminary GHX options and size/cost for feasibility analysis in PowerPoint format.

Pricing

Description	Rate/hr	Qty	Line Total
Energy Model	\$150.00	40	\$6,000.00
GLD Model & Review of Geology	\$175.00	8	\$1,400.00
Financial Analysis	\$175.00	12	\$2,100.00
Presentations & Meetings	\$175.00	12	\$2,100.00
Total (USD)			\$11,600.00

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